

# Yingxian Li . Jarvis Consulting

I recently graduated from University of Toronto with a degree in data analytics and have been building hands-on experience through data engineering projects at Jarvis Consulting Group. My background includes working with Python, SQL, PySpark, Databricks, PostgreSQL, and cloud-based data platforms to build analytics, backend, and distributed data processing solutions. What excites me most about the data industry is the opportunity to solve real business problems with technology and to keep learning new tools and systems. I am especially passionate about data engineering because it combines problem-solving, scalability, and practical business impact.

## Skills

**Proficient:** Python, Linux/Bash, RDBMS/SQL, Agile/Scrum, Git

**Competent:** SQL, Python, Pyspark, Databricks, PostgreSQL

**Familiar:** Java, Docker, Azure, Zeppelin, Maven

## Jarvis Projects

Project source code: [https://github.com/jarviscanada/jarvis\\_data\\_eng\\_YingxianLi](https://github.com/jarviscanada/jarvis_data_eng_YingxianLi)

### Cluster Monitor [GitHub]:

- Built a Linux-based cluster monitoring agent that collects host CPU/memory/disk metrics and stores them in PostgreSQL.
- Automated metric collection with Bash scripts and scheduled jobs, enabling near real-time visibility into server health.
- Wrote SQL queries to analyze resource usage trends and support capacity planning.

### RDBMS and SQL [GitHub]:

- Designed an ERD and implemented a relational schema using PostgreSQL DDL with primary/foreign keys and constraints.
- Wrote optimized SQL queries (joins, aggregates, subqueries, CTEs, window functions) to answer business questions.

**Core Java Apps [GitHub]:** Grep App: Built a Java CLI app that mimics Linux **grep** by recursively scanning directories, matching lines using regex (**Pattern**), and writing results to an output file; improved scalability by streaming file contents line-by-line instead of loading whole files.

### Python Data Analytics [GitHub]:

- Built a Python-based retail analytics proof of concept for the London Gift Shop (LGS) marketing team using Jupyter Notebook.
- Cleaned and transformed historical transaction data to support exploratory analysis and business insight generation.
- Generated marketing-focused insights to help identify high-value, loyal, inactive, and at-risk customers.

### Spark [GitHub]:

- Redesignated the earlier Python retail analytics workflow into a scalable Spark-based solution for larger datasets.
- Built the main implementation in Databricks on Azure using PySpark DataFrame APIs for data cleaning, transformation, aggregation, and business analysis.
- Analyzed retail transaction data to support use cases such as revenue calculation, monthly sales trends, and customer behavior analysis.

### Databricks [GitHub]:

- Built an end-to-end Databricks data pipeline using a medallion architecture, transforming raw data through Bronze, Silver, and Gold notebooks for analytics-ready reporting.
- Orchestrated the workflow with Databricks Jobs, sequencing notebook tasks and dashboard refresh steps to automate data processing from ingestion to visualization.
- Created interactive Databricks dashboards to present key business insights and KPIs from curated Gold-layer data for stakeholder reporting.

## Highlighted Projects

**Stock Analytics Pipeline:** Developed a medallion architecture pipeline in Databricks with DLT to ingest, clean, and transform stock API data across bronze, silver, and gold tables. Engineered analytical features such as rolling price change, percentage change, and volume trend metrics over 7/30/90-day windows for downstream reporting. Built interactive dashboard visualizations in Databricks SQL to monitor stock movement, compare tickers, and analyze trends with dynamic filters.

## Professional Experiences

**Data Engineering Trainee, Jarvis Consulting Group (2026.01 - Present):** Worked as a Data Engineering Trainee, building hands-on projects in Java, Python, SQL, Spark, and cloud/data platforms.

**Data Analyst, VS-Consulting Group (2022.05-2022.08):** Collected, cleaned, and transformed client data from multiple sources into structured datasets for analysis. Developed automated Excel VBA workflows and Power BI dashboards, reducing manual reporting effort and improving visibility into key business metrics. Built analytical models on selected KPIs to support data-driven decision-making aligned with stakeholder requirements. Collaborated with consultants and data engineers to deliver data solutions and present insights to project leads and clients.

## Education

**University of Toronto (2023-2025),** Master of Engineering, Data Analytics and Machine Learning

**University of Toronto (2019-2023),** Honours Bachelor of Science, Mathematics - Scholarship - University of Toronto Dean List (2020-2022) - GPA: 3.7/4.0

## Miscellaneous

- Databricks Certified Data Engineer Associate
- Microsoft Power BI Data Analyst (PL-300)
- SAS Base Programmer Certification
- SAS Advanced Programmer Certification
- Winner
- Basketball player
- Gym training